

Lab 8

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6B2

Example code

writer

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/stat.h>

int main()
{
    const char *fifo = "myfifo"; // Named pipe file

    // Create the FIFO (named pipe)
    mkfifo(fifo, 0666);

    // Open FIFO for writing
    int fd = open(fifo, O_WRONLY);

    // Redirect standard output to FIFO
    dup2(fd, 1);

    // Write message to FIFO
    printf("Hello from Writer Process!\n");

    close(fd);

    return 0;
}
```

reader

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    const char *fifo = "myfifo"; // Named pipe file
    char buffer[100];
```

```

// Open FIFO for reading
int fd = open(fifo, O_RDONLY);

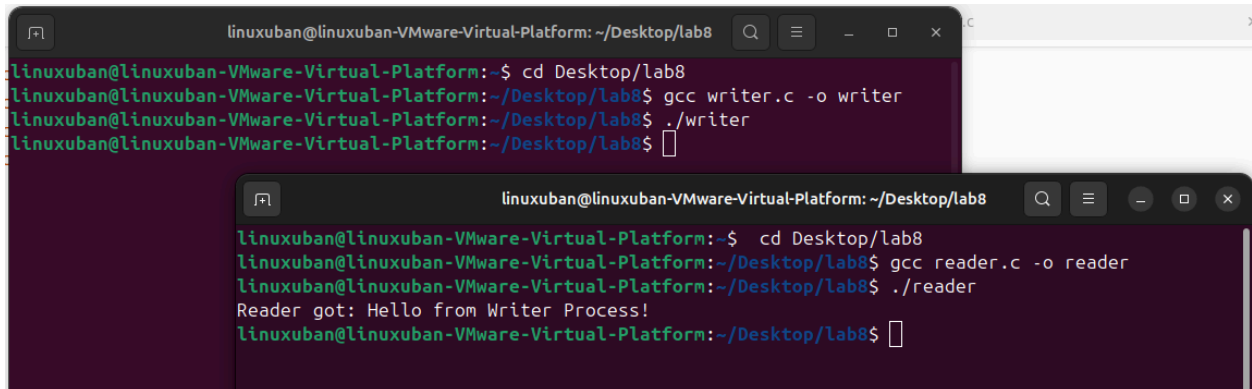
// Redirect standard input to FIFO
dup2(fd, 0);

// Read from FIFO and print
fgets(buffer, sizeof(buffer), stdin);
printf("Reader got: %s", buffer);

close(fd);

return 0;
}
// First run reader, then writer
//Run reader first, otherwise writer may block
//mkfifo creates a file named "myfifo" in your directory

```



The image shows two terminal windows from a Linux virtual machine. The top window shows the compilation of 'writer.c' to 'writer' and its execution. The bottom window shows the compilation of 'reader.c' to 'reader' and its execution, which outputs 'Reader got: Hello from Writer Process!'.

```

linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop/lab8
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$ cd Desktop/lab8
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$ gcc writer.c -o writer
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$ ./writer
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$

linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop/lab8
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$ gcc reader.c -o reader
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$ ./reader
Reader got: Hello from Writer Process!
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$

```

Q1

Server.cpp

```

#include <iostream>
#include <fcntl.h>
#include <unistd.h>
#include <sys/stat.h>

using namespace std;

```

```

int main()
{
    const char *fifo1 = "fifo1"; // Client → Server
    const char *fifo2 = "fifo2"; // Server → Client

    mkfifo(fifo1, 0666);
    mkfifo(fifo2, 0666);

    int fd_read = open(fifo1, O_RDONLY);
    int fd_write = open(fifo2, O_WRONLY);

    char op;
    int a, b, result;

    // Read input from client
    read(fd_read, &op, sizeof(op));
    read(fd_read, &a, sizeof(a));
    read(fd_read, &b, sizeof(b));

    // Perform operation
    switch(op)
    {
        case '+': result = a + b; break;
        case '-': result = a - b; break;
        case '*': result = a * b; break;
        case '/': result = a / b; break;
        default: result = 0;
    }

    // Send result to client
    write(fd_write, &result, sizeof(result));

    close(fd_read);
    close(fd_write);

    return 0;
}

```

Client.cpp

```

#include <iostream>
#include <fcntl.h>
#include <unistd.h>

using namespace std;

```

```

int main()
{
    const char *fifo1 = "fifo1"; // Client → Server
    const char *fifo2 = "fifo2"; // Server → Client

    char op;
    int a, b, result;

    cout << "Enter operator (+ - * /): ";
    cin >> op;

    cout << "Enter first number: ";
    cin >> a;

    cout << "Enter second number: ";
    cin >> b;

    int fd_write = open(fifo1, O_WRONLY);
    int fd_read = open(fifo2, O_RDONLY);

    // Send data to server
    write(fd_write, &op, sizeof(op));
    write(fd_write, &a, sizeof(a));
    write(fd_write, &b, sizeof(b));

    // Read result from server
    read(fd_read, &result, sizeof(result));

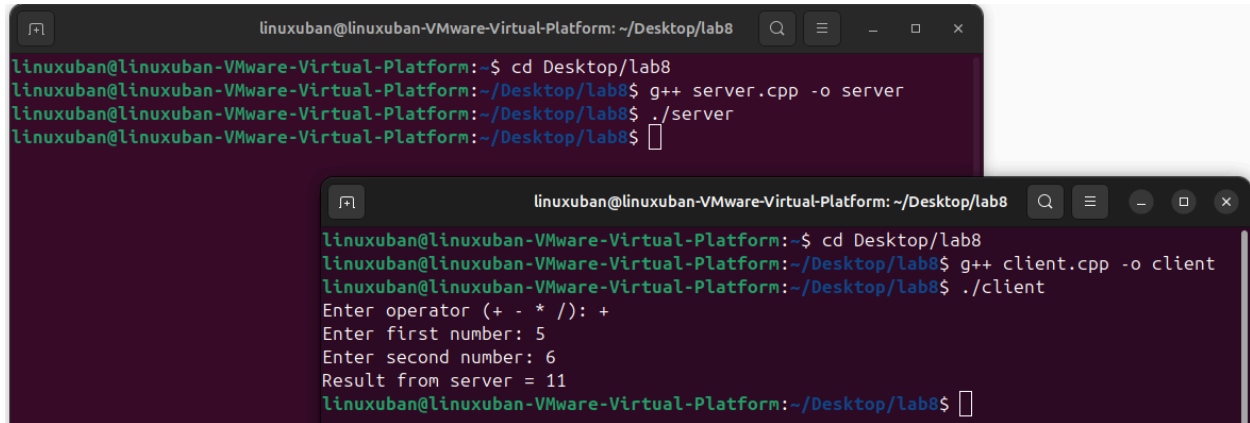
    cout << "Result from server = " << result << endl;

    close(fd_write);
    close(fd_read);

    return 0;
}

```

//Run server then client



```
linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop/lab8
linuxuban@linuxuban-VMware-Virtual-Platform:~$ cd Desktop/lab8
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$ g++ server.cpp -o server
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$ ./server
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$

linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8
linuxuban@linuxuban-VMware-Virtual-Platform:~$ cd Desktop/lab8
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$ g++ client.cpp -o client
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$ ./client
Enter operator (+ - * /): +
Enter first number: 5
Enter second number: 6
Result from server = 11
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$
```

Q2

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main()
{
    int fd[2];
    pipe(fd); // create pipe

    pid_t pid = fork();

    if (pid > 0)
    {
        // Parent → executes "ls"
        close(fd[0]); // close read end
        dup2(fd[1], 1); // stdout → pipe write end
        close(fd[1]);
        execlp("ls", "ls", NULL);
    }
    else if (pid == 0)
    {
        // Child → executes "sort"
        close(fd[1]); // close write end
        dup2(fd[0], 0); // stdin ← pipe read end
        close(fd[0]);
        execlp("sort", "sort", NULL);
    }

    return 0;
}
```

}

```
linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop/lab8
linuxuban@linuxuban-VMware-Virtual-Platform:~$ cd Desktop/lab8
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$ gcc q2.c -o q2
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$ ./q2
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$ client
client.cpp
fifo1
fifo2
myfifo
q2
q2.c
reader
reader.c
server
server.cpp
writer
writer.c

linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab8$
```